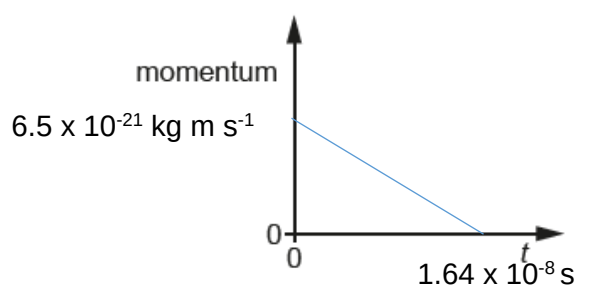
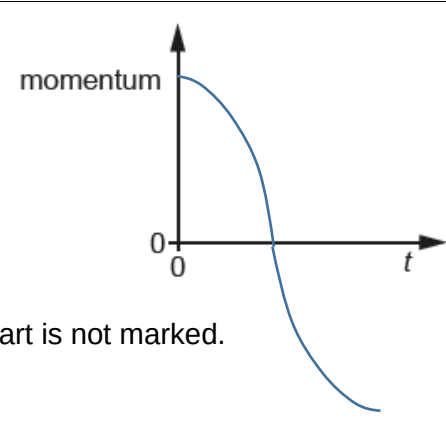


4	(a)	To the left or from B to A	B1
	(b)	$v^2 = u^2 + 2as$ $0^2 = (3.9 \times 10^6)^2 + 2a(0.032)$ M1 $a = -2.38 \times 10^{14}$ A1 Accept if student uses work done = loss of K.E	
	(c)	$F = Eq$ $ma = Eq$ $E = \frac{ma}{q} = \frac{1.67 \times 10^{-27} (2.38 \times 10^{14})}{1.6 \times 10^{-19}}$ C1 $E = 2.48 \times 10^6$ A1 Accept if student uses $qEd = \text{loss in K.E}$	
	(d)	 B2 for correct axis labels (one for momentum, one for time) B1 for shape	
	(e)	(ii) The proton will not reach point B/come to rest before point B for Fig 4.3. and subsequently accelerate in the opposite direction OR Momentum/velocity for proton is lower at point A for Fig 4.3. OR The deceleration for proton is not constant for Fig 4.3 (it experiences increasing deceleration in the direction of travel.)	B1
	(ii)	 The negative part is not marked. The gradient of the graph gives resultant force (electric force) and it increases closer to the charge.	

			B1 for shape (note: the gradient at time = 0 cannot be horizontal and the gradient at momentum = 0 cannot be vertical)	
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5	(a)	(i)	The magnetic flux through a plane surface is the product of the flux	B1
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